

AI-Powered Rate Confirmation Processing Automation

UiPath • Document Understanding • AI Workflow Automation

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Business Context



In carrier operations, rate confirmations are received daily from brokers in varying formats and must be processed quickly to support reporting, profitability analysis, and operational decision-making.

Operational Challenges

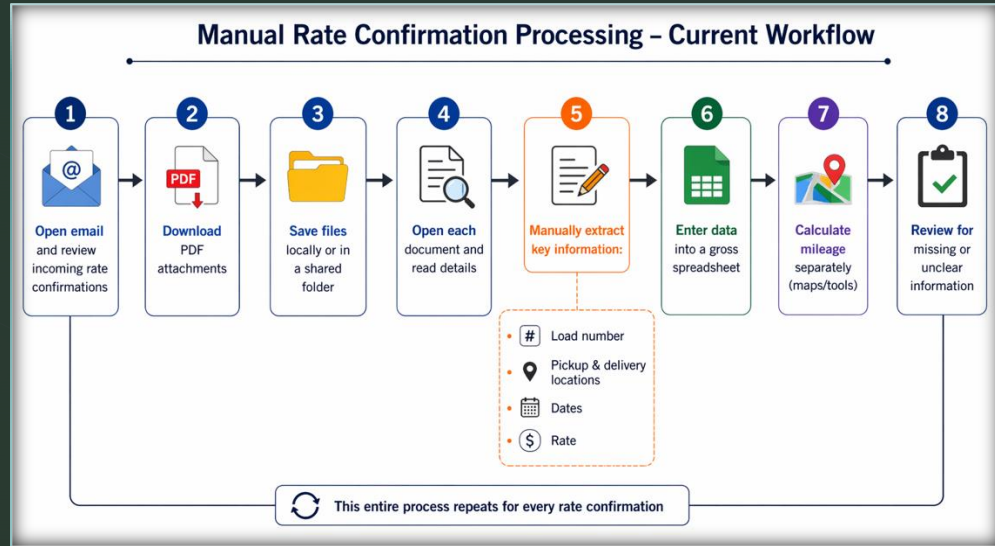
- High daily volume of rate confirmations
- Manual shipment data extraction
- Separate mileage lookup process
- Repetitive spreadsheet entry
- Inconsistent broker document formats
- Difficult to scale reporting workflows

Business Impact

- ~5 hours/week spent on repetitive processing
- Increased risk of manual data errors
- Delayed RPM and profitability reporting
- Reduced operational visibility and scalability

Manual Rate Confirmation Processing

The current manual workflow for processing rate confirmations involves repetitive tasks such as document handling, manual data extraction, and separate mileage calculation using external tools like Google Maps, where each location must be entered manually.



Automation Solution

An end-to-end UiPath automation was developed to streamline rate confirmation processing by combining AI-powered document extraction, mileage enrichment, validation logic, and structured spreadsheet reporting into a single scalable workflow.

The solution reduces repetitive manual processing while improving operational visibility, reporting efficiency, and data consistency.

Core Capabilities:

- Automatically retrieves rate confirmations from email
- Stores and organizes documents in Google Drive
- Extracts shipment data using Document Understanding
- Calculates mileage using AI-based enrichment
- Writes structured output directly into reporting spreadsheets
- Flags incomplete or low-confidence data for manual review
- Sends automated validation alerts when issues are detected

Proposed Automation Solution

This UiPath automation handles the end-to-end processing of rate confirmations, from email retrieval to data extraction, AI-based mileage calculation, and structured output, with built-in validation and error handling.



Automation Architecture Components

The automation was implemented in UiPath by combining multiple components, including email integration, file handling, document understanding, AI-based enrichment, and validation logic.

Key Components:

- Email integration to retrieve rate confirmations
- File storage and management using Google Drive
- Document Understanding pipeline for data extraction
- AI integration for mileage calculation
- Validation logic for data quality control
- Spreadsheet integration for structured output

Document Understanding & Custom Extractor

A custom generative extractor was configured to extract key shipment data from rate confirmations, which often vary in format across different brokers.

Key Points:

- Uses Document Understanding to process unstructured PDF documents
- Custom fields defined for logistics-specific data:
- Load number
- Pickup & delivery locations
- Dates
- Rate
- Generative extractor used for flexibility across different formats
- Designed to handle varying document quality and layouts

Custom extractor enables accurate data extraction even from non-standard documents.

Field name	Instruction
Name of the field to extract from the document	Instruction about what information to extract about the field
Load number	Extract the load number from the rate confirmation. This is the broker reference identifying the load. Return only the load number as written in
Pickup location	Extract the pickup location for the load. Return city and state only, and zip code if available.
Delivery location	Extract the delivery location for the load. Return city and state only, and zip code if available.
Pickup date	Extract the scheduled pickup date for the load. Return date in mm/dd/yy format.
Total rate	Extract the total agreed carrier pay or total rate for this load. Return numbers only in 0.00 format without currency symbol or commas.

 **Extract Key Shipment Data**

 DU models not currently defined as part of Solution definitions. Please make sure they are created manually and exist when deploying the solution to other tenants.

Input*

Document Understanding project *

Extractor *

Document Type Details*

AI-Powered Mileage Calculation

LLM-based mileage estimation was implemented as a lightweight AI enrichment approach during initial workflow prototyping.

Key Points:

- Uses pickup and delivery locations extracted from documents
- Sends structured prompt to LLM
- Returns estimated mileage for each trip
- Eliminates manual lookup in tools like Google Maps
- Ensures consistent and fast calculation

AI transforms extracted location data into actionable operational metrics.

 **Calculate Mileage Using AI** 

 Indexes used in these activities are not currently defined as part of Solution definitions. Please make sure they are created manually and exist when deploying the solution to other tenants.

UiPath GenAI Activities connection

 UiPath GenAI Activities 

Model name

anthropic.claude-haiku-4-5-20251001-v1:0 (Claude)

Prompt *

You are helping a logistic manager to calculate the miles for each trip. Look at the pickup  documentData.Data.PickupLocation and delivery  documentData.Data.DeliveryLocation locations to calculate the total mileage. The answer must be numbers only.

Data Mapping & Debugging

Extracted and enriched data is mapped into a structured format before being written to the spreadsheet, with logging used to verify results and simplify debugging.

Key Points:

- Extracted fields are mapped to structured output columns in the spreadsheet
- Ensures consistent data format for reporting and analysis
- Log messages used to verify extracted values before writing output
- Helps quickly identify missing or incorrect data
- Simplifies debugging and improves reliability of the workflow

Logging and structured mapping ensure transparency and make the automation easier to maintain and troubleshoot.

Data Mapping & Debugging

Data is mapped into a structured output format, with logging implemented to validate extracted values and ensure easier debugging.

 Validate Extracted Data 

Message *

Load #: {x} documentData.Data.LoadNumber (confidence: {x} documentData.Data.LoadNumber.Confidence)

Broker name: {x} documentData.Data.BrokerName (confidence: {x} documentData.Data.BrokerName.Confidence)

Log Level

Select a level or leave empty for info 

 Write Processed Data to Spreadsheet 

Google Sheets connection

 @gmail.com_2 X

Spreadsheet* [Use spreadsheet template](#)

Maryna Khlynina gross 

Range *

'total gross' (Sheet) X 

How to write

Append 

Has headers

☒ True ☐ False 

What to write

Individual Fields 

Load #

{x} documentData.Data.LoadNumber @ 

Broker name

{x} documentData.Data.BrokerName @ 

Shipping date

{x} documentData.Data.PickupDate @ 

Validation Logic

Validation logic ensures that only reliable and complete data is processed, while identifying and flagging potential issues.

Key Points:

- Identifies missing or not extracted fields (confidence = -1)
- Recognizes that custom extractor fields return confidence = 0 by default
- Flags incomplete or suspicious data based on validation rules
- Sends email notifications for manual review
- Prevents incorrect or incomplete data from entering the system

Built-in validation ensures data accuracy and reliability before final output.

In this solution, confidence scores are interpreted differently due to the use of a custom extractor. A value of -1 indicates missing data, while 0 represents extracted data without a confidence model.

The screenshot shows a form titled "Send Notification for Manual Review". It includes a "Gmail connection" field with a dropdown menu showing an email address. Below this is a "Save as draft" section with radio buttons for "True" and "False", where "False" is selected. The "To*" field contains an email address. The "Subject" field is labeled "Rate confirmation requires manual check". The "Body" field contains a rich text editor with the text "Rate confirmation requires manual check! Low confidence on rate confirmation: {x} CurrentItem {x} CurrentItem.Url".

The screenshot shows a configuration for a "Check Data Quality" rule. The "Condition *" field contains the expression "documentData.Data.LoadNumber.Confidence < 0 OR documentD...". Below this is a "Then" section with a "Log Data Quality Issue" action. The "Message *" field contains the text "Rate confirmation requires manual check: low confidence {x} CurrentItem". The "Log Level" field is set to "Warn".

Key Outcomes

This automation successfully processes rate confirmations end-to-end, extracting, enriching, and structuring data into a usable format.

Key Outcomes:

- ~70–80% reduction in manual processing
- Successful mileage calculation using AI
- Automated dozens of confirmations
- Reduced repetitive data entry
- Faster RPM reporting
- Flags low-confidence data for review

The automation reliably transforms unstructured documents into structured, actionable data.

Structured Data Output

A	B	C	D	E	F	G	H	I
Load #	Broker name	Shipping date	Pickup location	Delivery date	Delivery location	Total rate	Miles	RPM
306837	PLS LOGISTICS SERVICES	02/08/23	Botkins, OH 45306	02/10/23	orlando fl 32824	\$3,600.00	1087	
451400H	International Transport Systems, Inc.	03/14/23	LA VERGNE, TN 37086	03/15/23	Pine Bluff, AR 71602	\$1,150.00	289	
CPL808978633	Custom Pro Logistics	01/28/20	Horse Cave, KY 42749	01/29/20	OCONOMOWOC, WI 53066	\$1,300.00	628	
S-0419-1725	HGR Industrial Surplus	01/15/20	Montgomery, IL 60538	01/16/20	Euclid, OH 44117	\$1,300.00	651	
3440KJ	Beard Transport	03/02/23	Blackshear, GA, 31516	03/04/23	Canterbury, NH, 03224	\$3,500.00	1247	
	MAISON LUXE	03/13/23	Upper Sandusky, OH 43351	03/14/23	Ashland City, TN 37015	\$1,500.00	534	
	ADM LOGISTICS, INC.	01/27/20	DES PLAINES IL 60018	01/28/20	NASHVILLE TN 37210	\$1,650.00	512	
505684E	Transplace Texas, LP	01/28/20	ALSIP, IL 60803	01/30/20	LOXLEY, AL 36551	\$1,800.00	1047	
3690729	MERCER TRANSPORTATION CO.	03/13/23	Upper Sandusky, OH 43351	03/14/23	ASHLAND CITY, TN 37015	\$1,623.73	592	
C-213547	Coyote Logistics, LLC	11/12/20	Sterling Heights, MI 48314	11/12/20 - 11/13/	Bellevue, OH 44811	\$1,500.00	286	

Processed data is organized into a structured spreadsheet, making it easy to spot missing information and ensure data completeness. With mileage already calculated, RPM can be derived instantly using a single formula applied across all entries, enabling faster and more scalable performance analysis.

Validation & Monitoring Output

Confidence scores appear as 0 for custom extractor fields and -1 when no value is detected.

An automated alert triggered by validation logic when incomplete or missing data is identified in the extracted results.

 11:10:52 PM Load #: (confidence: -1)
Broker name: ADM LOGISTICS, INC...
Pick up date: 01/27/20 (confidence:...
Pick up: DES PLAINES IL 60018 (con...
Delivery date: 01/28/20 (confidenc...
Delivery: NASHVILLE TN 37210 (con...
Rate: 1650.00 (confidence: 0)

 11:10:53 PM mileage: 520

 11:10:53 PM Rate confirmation requires manua...

 11:11:10 PM Load #: 0505637 (confidence: 0)
Broker name: Transplace Texas, LP ...
Pick up date: 01/28/20 (confidence:...
Pick up: ALSIP, IL 60803 (confidenc...
Delivery date: 01/30/20 (confidenc...
Delivery: LOXLEY, AL 36551 (confid...
Rate: 1800.00 (confidence: 0)

 11:11:11 PM mileage: 1047

Rate confirmation requires manual check 

 1@gmail.com>
to me ▾

11:10 PM (19 minutes ago) ☆ ☺

Rate confirmation requires manual check! Low confidence on rate confirmation: UiPath.GSuite.Drive.Models.GDriveRemoteItemhttps://drive.google.com/uc?id=1ai60HPbcABBHqV_ynExiqQdxAbQa1_E3&export=download

One attachment • Scanned by Gmail  Added to Drive  Compress PDF



Business Impact

Operational Improvements:

- Reduced repetitive manual work
- Improved reporting efficiency
- Increased data consistency
- Enabled scalable processing workflow

Technical Achievements:

- AI-powered document extraction
- Automated mileage enrichment
- Validation & monitoring logic
- Structured spreadsheet output

Production Readiness & Future Enhancements

How AI improved it:

- Automated extraction of key shipment data from unstructured documents
- AI-based mileage calculation eliminates the need for manual lookup
- Enhances scalability by handling large volumes of documents efficiently

Future Enhancements:

- Replace LLM mileage with Google Maps API for higher accuracy
- Implement a custom-trained extraction model to improve confidence scoring and document accuracy across varying broker formats

Business Value

- **Time Efficiency:**
Automates repetitive tasks such as document review, data entry, and mileage calculation
- **Accuracy & Data Quality:**
Reduces human errors and flags missing or incomplete data for review
- **Scalability:**
Enables processing of large volumes of rate confirmations without additional manual effort
- **Operational Visibility:**
Structured data allows instant calculation of KPIs like RPM and overall performance
- **Faster Decision-Making:**
Provides clean, ready-to-use data for analyzing profitability and load efficiency